

Model Card

Model Name: pv_detector.pt

Version: v1.0

Model Architecture: UNet++ with EfficientNet-B3 encoder

Deployment Type: Batch + human-in-the-loop review via Streamlit application

Primary Use Case: Governance-grade remote verification of rooftop solar PV installations

1. Model Overview

This system detects and segments rooftop photovoltaic (PV) panels from high-resolution satellite imagery and produces **audit-ready outputs** for subsidy verification workflows.

The model is integrated into a **controlled end-to-end pipeline** that includes:

- Deterministic preprocessing
- Calibrated inference
- Spatial buffering logic
- Quality control (QC) gating
- Optional human review

Primary outputs per site:

- **has_solar** (boolean)
- **confidence** (0–1)
- **pv_area_sqm_est**
- **buffer_radius_sqft** (1200 or 2400)
- **qc_status** (VERIFIABLE / REVIEW / LOW_CONFIDENCE / NOT_VERIFIABLE)
- Polygon mask or fallback raster mask
- Visual audit overlay

Intended use

- Remote verification of rooftop solar installations for governance and subsidy auditing
- Decision support for DISCOMs and public agencies

Out-of-scope use

- Energy generation estimation
- Structural safety assessment
- Legal ownership or dispute resolution
- Real-time or continuous monitoring

2. Data Used

Training Dataset

Jiang, Hou, Ling & Yujun (2021)

Multi-resolution dataset for photovoltaic panel segmentation from satellite and aerial imagery

Zenodo, v1.0

DOI: <https://doi.org/10.5281/zenodo.5171712>

Dataset characteristics

- RGB satellite and aerial imagery
- Pixel-level PV segmentation masks
- Multiple spatial resolutions
- Diverse panel orientations and roof materials

Data limitations

- Geographic bias toward regions represented in the dataset
- Limited representation of Indian rural roof typologies
- Fewer samples with dense rooftop clutter (tanks, vegetation)
- No explicit temporal diversity guarantees

These limitations directly influence QC design and retraining guidance.

3. Model Architecture & Inference Logic

Architecture

- **Encoder:** EfficientNet-B3 (ImageNet pretrained)
- **Decoder:** UNet++ with nested skip connections
- **Output:** Single-channel PV probability map

Rationale

- UNet++ improves multi-scale feature fusion, critical for detecting small or fragmented PV arrays
- EfficientNet-B3 balances accuracy and deployment efficiency

Inference Pipeline (as implemented)

1. Satellite image fetched via Google Static Maps API
2. Canonical preprocessing applied:
 - RGB enforcement
 - EXIF rotation correction
 - Aspect-ratio preserving resize
 - Center padding to **640×640**
 - ImageNet normalization
3. Model inference → logits
4. Temperature-scaled sigmoid ($T = 1.5$)
5. Binary mask via threshold = 0.5
6. Mask resized back to original image resolution

7. Confidence computed using:
 - Mean probability inside predicted mask
 - 95th percentile probability
8. Spatial filtering using circular buffers (1200 / 2400 sqft)
9. Area estimation via GSD-based pixel-to-meter conversion

4. Assumptions

The system assumes:

- Rooftop PV panels are visually distinguishable in RGB imagery
- Input imagery resolution is sufficient at zoom level 20
- Panels are not fully occluded by shadows, trees, or rooftop equipment
- Imagery is recent enough to reflect current rooftop state
- Geographic coordinates may contain minor jitter

Violation of these assumptions triggers conservative QC behavior.

5. Quality Control & Decision Logic

Buffer-based reasoning

- Primary verification within **1200 sqft** radius
- Secondary verification within **2400 sqft** radius if no PV detected in 1200 sqft
- Optional controlled extension beyond 2400 sqft when partial detection exists

QC Status Rules

- **VERIFIABLE**
 - PV area $> 1.0 \text{ m}^2$

- Confidence > 0.8
- **REVIEW**
 - Confidence between 0.4 and 0.8
- **LOW_CONFIDENCE**
 - Confidence < 0.4
- **NOT_VERIFIABLE**
 - Missing imagery
 - Model unavailable
 - Severe preprocessing or inference failure

Human-in-the-loop

- “Smart Review” mode automatically routes low-confidence cases for manual validation
- Final approval always produces a traceable overlay artifact

6. Known Limitations & Biases

- Reduced performance under heavy shadows or cloud cover
- False positives from visually similar objects:
 - Skylights
 - Solar water heaters
 - Dark rooftop materials
- False negatives for:
 - Small or partially occluded installations
 - Panels aligned with roof color
- Urban bias due to training data composition

- Sensitivity to stale imagery

These limitations are explicitly surfaced through QC status rather than hidden.

7. Failure Modes

Common failure cases include:

- Low-resolution or blurred imagery
- Dense rooftop clutter (tanks, vegetation)
- Partial PV arrays extending outside buffer boundaries
- Missing API connectivity
- Model file unavailability
- Spatially arranged PV

In such cases, the system defaults to **NOT_VERIFIABLE** or **REVIEW**, avoiding overconfident predictions.

8. Evaluation Summary

Model performance is evaluated using:

- **F1 score** for PV presence detection
- **IoU** for segmentation quality
- **RMSE** for PV area estimation (m²)

Metrics are reported alongside QC status and confidence to support auditability rather than raw automation.